FINASHEALTH

PROJECT DOCUMENTATION

AI-Powered Credit Health & Financial Wellness Platform for India

CREDIT ASSISTANT

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Live Website: https://finashealth.web.app/

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1. Project Overview

1.1 Introduction

FinasHealth, also described as Credit Assistant, is an AI-powered credit-health and financial-wellness platform designed for Indian retail borrowers. It aims to help users understand important aspects of their credit profile and personal financial position through a single web-based experience. The project focuses on credit awareness, financial indicators, guided insights and practical steps that can support healthier borrowing behaviour.

The platform brings together common credit-health measures such as a reported CIBIL score, debt-to-income ratio (DTI), credit utilisation and savings-related indicators. It combines a ten-step onboarding experience, dashboard charts, a what-if simulator and an AI-assisted advisor. The advisor is designed to convert the user's financial inputs into understandable observations and a structured five-step improvement roadmap.

FinasHealth is an educational and informational platform. It does not itself issue an official credit-bureau score, make lender decisions, guarantee loan approval or replace qualified financial advice.

1.2 Scope of the System

The system is an India-focused financial wellness and credit-management web application for retail borrowers. It is intended for people who want to understand their credit health, monitor relevant financial indicators and learn how financial choices may affect their overall profile. The documented capabilities include guided onboarding, account authentication, dashboard visualisations, a what-if simulator and an AI advisor powered by Gemini 2.5 Flash with a deterministic fallback.

The scope is credit-health education and personal financial awareness. FinasHealth is not a payment gateway or payment-processing application. The platform works with user-provided or reported information; any direct connection to credit bureaus, banks or lenders must be treated as future scope unless verified in the implementation.

1.3 Project at a Glance

Attribute	Detail
Project name	FinasHealth
Product concept	Credit Assistant
Domain	Personal finance, credit health and financial wellness
Geographic focus	India
Target users	Retail borrowers, salaried individuals, students, self-employed and credit-card users
Credit-score context	Reported CIBIL score range: 300–900
Key indicators	DTI, credit utilisation and savings-related measures
Core modules	Ten-step onboarding, dashboard, what-if simulator and AI Advisor
AI engine	Gemini 2.5 Flash with deterministic fallback
Frontend	React 19, Vite, Tailwind CSS and Recharts
Backend	Node.js / Express runner and FastAPI / Pydantic services
Data access and storage	SQLAlchemy and SQLite
Security	JWT authentication and PBKDF2 password hashing

Live website	https://finashealth.web.app/

Table 1.1: FinasHealth project summary

2. Problem Statement

2.1 Problem Description

Many individuals use credit cards, personal loans, consumer loans or other borrowing facilities without fully understanding the factors that influence their credit health. A credit score is only one part of a wider financial picture. Debt obligations, credit-limit usage, repayment behaviour and the ability to manage existing commitments are all relevant to financial wellness.

Users may find it difficult to interpret a reported credit score, calculate their debt-to-income ratio, understand credit utilisation or decide which practical action to take first. Information is often spread across different financial websites and calculators. This can make it harder for students and first-time borrowers to connect individual indicators with a clear improvement plan.

FinasHealth addresses this challenge by bringing key credit-health indicators, user-provided financial information, explanatory insights and AI-assisted guidance into one accessible platform. It is designed to make financial information easier to understand and encourage informed, responsible financial habits.

2.2 Challenges Addressed

- Credit-score awareness: users may know their reported score but may not understand the 300–900 range or the factors that can influence it.
- Debt burden visibility: users may not know what proportion of monthly income is committed to debt repayments.
- Credit-limit management: credit-card utilisation can be difficult to assess without comparing used credit with the total available limit.
- Fragmented information: users may need several tools to understand different parts of their financial position.
- Action prioritisation: generic advice may not provide a clear sequence of practical steps.
- Scenario exploration: users may want to understand how hypothetical changes in debt or credit balances affect calculated indicators.

3. Objectives

3.1 Primary Objectives

1. Provide a guided onboarding process to collect the financial information required by the platform.
2. Help users record or review their reported credit score within the Indian credit-score context.
3. Calculate and explain debt-to-income ratio using monthly debt obligations and income.
4. Calculate credit utilisation using used credit and total available credit limit.
5. Present credit-health and financial-wellness indicators through a consolidated dashboard.
6. Provide a what-if simulator to explore hypothetical changes in selected financial inputs.
7. Generate AI-assisted insights and a structured five-step improvement roadmap using Gemini 2.5 Flash.
8. Provide a deterministic fallback when the AI service is unavailable or returns an unusable response.
9. Protect account access through authentication and password-hashing mechanisms.
10. Maintain an extensible architecture for future credit-health and financial-wellness features.

3.2 Objective-to-Capability Mapping

Objective	Supporting capability
Guided financial profile	Ten-step onboarding flow
Credit awareness	Reported score entry and 300–900 score context
Debt burden understanding	DTI calculation and explanation
Credit-limit awareness	Credit utilisation calculation
Centralised overview	Dashboard cards and charts
Decision exploration	What-if simulator
Educational guidance	Gemini 2.5 Flash advisor and five-step roadmap
Service continuity	Deterministic AI fallback
Account protection	JWT authentication and PBKDF2 password hashing

4. Proposed Solution

4.1 Solution Description

The proposed solution is a web application that guides users through the collection of relevant financial information and converts those inputs into understandable credit-health indicators. Users begin with account access and onboarding, proceed to the dashboard, and can explore calculations, scenario simulations and AI-assisted guidance.

The frontend is built with React 19, Vite, Tailwind CSS and Recharts. The documented backend architecture includes a Node.js/Express runner and FastAPI services using Pydantic and SQLAlchemy, with SQLite for relational data storage. The AI advisor communicates with Gemini 2.5 Flash to generate explanatory insights and a five-step roadmap. A deterministic fallback supports continuity if the external AI service is unavailable.

4.2 Functional Modules

Module	Purpose and role
Authentication	Provides account access using JWT-based authentication; passwords use PBKDF2 hashing.
Ten-step onboarding	Collects user profile and financial inputs for the credit-health view.
Dashboard	Presents a consolidated view of reported credit and calculated financial indicators.
Credit profile	Displays the reported credit score and relevant credit-health context.
DTI calculator	Calculates debt obligations as a percentage of monthly income.
Credit utilisation	Calculates the share of available revolving credit currently used.
What-if simulator	Allows users to test hypothetical changes and review recalculated indicators.
AI Advisor	Uses Gemini 2.5 Flash for insights and a five-step roadmap, with deterministic fallback.
Educational disclaimer	Clarifies that results are informational and not a lender decision or professional advice.

4.3 Implemented and Future Scope

Status	Capability
Documented current capabilities	Ten-step onboarding; authentication; dashboard charts; reported CIBIL score context; DTI and credit-utilisation calculations; savings indicators; what-if simulator; Gemini 2.5 Flash advisor; five-step roadmap; deterministic fallback; educational disclaimer.
Proposed / verify before claiming	Direct bureau integration; bank-account aggregation; live lender offers; payment processing; predictive score modelling; advanced product recommendations; expanded analytics.

FinasHealth is a financial-wellness and credit-management website, not a payment-processing application. External bureau or bank connections should be described as future scope unless they are implemented and verified.

5. Project Scenarios

The following representative scenarios illustrate how different users may benefit from the same credit-health tools. They describe intended use cases and do not imply that the platform independently verifies financial records.

5.1 Scenario 1: Salaried Professional

A salaried employee enters monthly income, recurring debt obligations, credit-card limits and outstanding balances during onboarding. The dashboard presents the reported score and calculated indicators. DTI helps the user understand the share of income committed to debt repayments, while credit utilisation highlights how much available revolving credit is being used. The AI advisor can translate these inputs into general improvement priorities.

5.2 Scenario 2: College Student / First-Time Credit User

A student or first-time credit user may have limited credit history and little familiarity with credit terminology. The platform can explain the meaning of a reported score, the importance of responsible repayment and the calculation of credit utilisation. The guided onboarding and educational insights can help users develop awareness before taking on additional borrowing.

5.3 Scenario 3: Self-Employed Borrower

A self-employed individual may have variable monthly income and multiple financial commitments. By entering representative income and debt information, the user can review the calculated DTI indicator and explore hypothetical changes through the simulator. Results should be interpreted in light of income variability; one month's information may not represent the full financial picture.

5.4 Scenario 4: Existing Credit-Card User

A credit-card user enters the total available credit limit and current used or outstanding credit. The platform calculates utilisation and presents it as an awareness indicator. The user can test a hypothetical reduction in outstanding balance and compare the resulting utilisation percentage. This explains the calculation without promising a particular credit-score movement.

5.5 Scenario Comparison

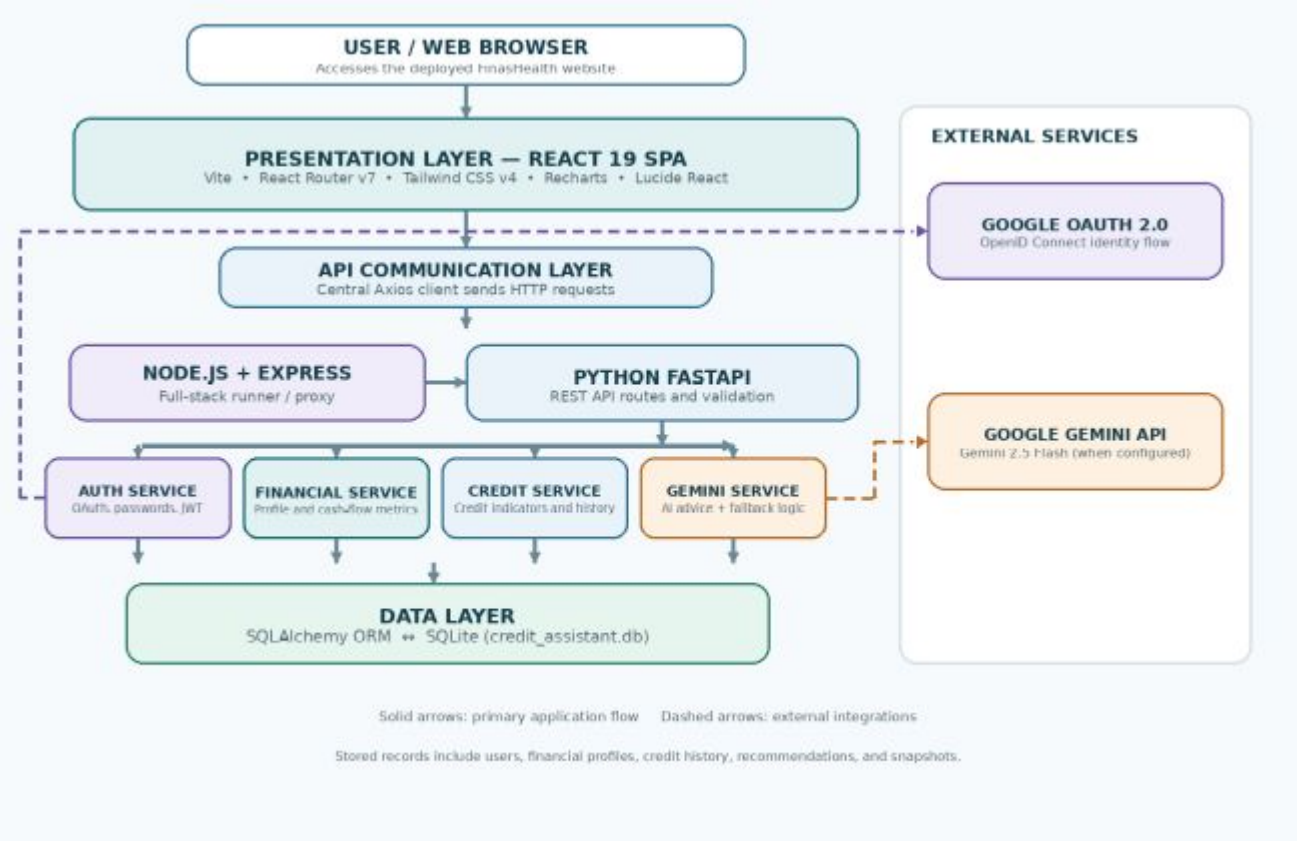
Aspect	Salaried	Student / new user	Self-employed	Credit-card user
Typical inputs	Income and EMIs	Profile and credit details	Variable income and debt	Limit and used amount
Main concern	Debt burden	Credit awareness	Changing affordability	Utilisation
Useful feature	DTI / dashboard	Onboarding / guidance	What-if simulator	Utilisation calculator
Benefit	Understand obligations	Build informed habits	Explore scenarios	Understand credit usage

6. System Workflow and Architecture

6.1 Architectural Overview

FinasHealth follows a layered web-application architecture. The presentation layer uses React 19 and Vite, Tailwind CSS for styling and Recharts for data visualisation. The documented application layer includes a Node.js/Express runner and FastAPI backend services. FastAPI uses Pydantic for request and data validation and SQLAlchemy for database access. SQLite provides relational data storage in the documented project setup. The AI integration communicates with the Gemini API for generated guidance.

JWT supports token-based authentication and PBKDF2 is used for password hashing. The architecture separates user-interface responsibilities, API handling, business logic, data access and AI-assisted guidance.



6.2 Component Responsibilities

Component	Responsibility
React 19	Builds the interactive user interface.
Vite	Provides frontend development and build tooling.
Tailwind CSS	Supports styling and responsive layout.
Recharts	Displays indicators through charts and visual summaries.
Node.js / Express	Acts as the application runner / API coordination layer.
FastAPI	Exposes backend endpoints and coordinates services.
Pydantic	Validates and structures API request and response data.
SQLAlchemy	Provides ORM and database access.
SQLite	Stores application data in relational form.
JWT / PBKDF2	Support token authentication and password hashing.
Gemini 2.5 Flash	Generates AI-assisted insights and the five-step roadmap.

6.3 Logical Architecture

Layer	Logical flow / components
Presentation	Browser → React 19 + Vite → Tailwind interface / Recharts dashboard
Application API	Frontend requests → Node.js / Express runner → FastAPI endpoints
Business services	Pydantic validation → credit calculations / simulator → advisor orchestration
Data	Service layer → SQLAlchemy → SQLite
AI	Advisor service → Gemini 2.5 Flash → response; deterministic fallback if unavailable
Security	JWT authentication; PBKDF2 password hashing

Figure 6.1: Logical architecture based on the documented project stack

6.4 Workflow

11. The user opens FinasHealth and registers or signs in.
12. The user completes the ten-step onboarding process and provides profile and financial inputs.
13. The frontend sends relevant information to backend services through the application API layer.
14. The backend validates the incoming data and processes the required credit-health calculations.
15. The dashboard presents reported score context and calculated indicators such as DTI and utilisation.
16. The user may change supported assumptions in the what-if simulator to explore hypothetical outcomes.
17. The AI advisor sends relevant context to Gemini 2.5 Flash and prepares insights and a five-step roadmap.
18. If the AI service cannot provide a usable response, the deterministic fallback is used where configured.

19. The user reviews the output with the educational disclaimer in mind.

7. Technology Stack

7.1 Technologies Used

Technology	Category	Role in project
React 19	Frontend library	Component-based interactive screens
Vite	Frontend tooling	Development server and build tooling
Tailwind CSS	CSS framework	Styling and responsive layout
Recharts	Chart library	Financial and credit-health visualisation
Node.js	JavaScript runtime	Runs the Express application layer
Express	Backend framework	Coordinates application requests
Python	Backend language	Supports FastAPI services
FastAPI	API framework	Defines backend endpoints
Pydantic	Data validation	Validates structured API data
SQLAlchemy	ORM	Manages database access
SQLite	Database	Relational data storage
Gemini 2.5 Flash	AI model	Generates insights and roadmap
JWT	Authentication	Token-based access
PBKDF2	Password hashing	Protects stored password representations

7.2 Hardware Requirements

Component	Suggested requirement
Processor	Modern dual-core processor or better
Memory	4 GB RAM minimum; 8 GB recommended for development
Storage	Free space for source code, dependencies and local database
Internet	Required for hosted access and Gemini API use
Display	Mobile, tablet or desktop display with a modern browser

These are practical recommendations, not measured minimum requirements from a formal performance test.

7.3 Software Requirements

Component	Requirement
Operating system	Windows, macOS or Linux
Browser	Current Chrome, Firefox, Edge or another modern browser
Frontend tools	Node.js and npm
Backend tools	Python environment for FastAPI
Code editor	Visual Studio Code or compatible IDE
Version control	Git and GitHub, if used in the project

AI service	Gemini API access and configured credentials
Database	SQLite and compatible SQLAlchemy configuration

8. Database Design

8.1 Database Technology

The documented backend uses SQLAlchemy as the database access layer and SQLite as the relational database. SQLAlchemy provides an abstraction between application services and the underlying database, allowing backend code to work with structured entities and queries. SQLite is suitable for a compact project setup and local development because it stores relational data in a database file.

Exact table names, column definitions, relationships and production database configuration must be verified from the current source-code models and deployment settings. This report therefore describes logical information groups rather than inventing physical schema details.

8.2 Data Elements Evident from the Project

Data group	Information represented	Purpose
User account	Account identity and authentication-related fields	Account access control
User profile / onboarding	Profile and financial context entered by the user	Provides calculation and advisor context
Credit profile	Reported score and relevant credit details	Credit-health presentation
Debt obligations	Monthly debt payment amounts	DTI calculation
Credit facilities	Available credit limit and used / outstanding amount	Utilisation calculation
Savings indicators	Income, expense or savings-related values	Financial-wellness summary
Simulation inputs	Hypothetical values entered in the simulator	Scenario comparison
Advisor context	Selected profile details and calculated indicators	AI-generated guidance

8.3 Logical Data Model

Conceptually, a user account is associated with a profile and financial information entered during onboarding. Credit, debt, utilisation and savings inputs are processed by backend services to produce calculated indicators. Simulation inputs are evaluated as hypothetical scenarios. The AI advisor receives selected context and returns guidance for presentation in the user interface.

Logical entity / service	Conceptual relationship
User	Owns or is associated with a profile.
Profile	Contains onboarding information supplied by the user.
Credit information	Provides reported score and credit-facility details.
Debt information	Provides monthly obligations for DTI.
Calculation service	Processes inputs into DTI and utilisation indicators.
Simulation service	Evaluates hypothetical values separately from actual records.
Advisor service	Requests AI guidance and prepares fallback output.

Figure 8.1: Conceptual data relationships; confirm the physical schema from source code

For final technical submission, insert an entity-relationship diagram generated from the actual SQLAlchemy models. This conceptual description is not a confirmed database schema.

9. Credit Calculations and AI Logic

9.1 CIBIL Score Context

FinasHealth is designed for the Indian credit ecosystem and uses the familiar CIBIL score range of 300 to 900 as a reference for credit-score awareness. A user-provided or reported score can be displayed and discussed in context. The value shown should be understood as information entered or supplied to the application; it should not be represented as a live bureau retrieval unless such integration is implemented.

A credit score is one indicator of credit history and repayment behaviour. Lenders may consider additional information and apply their own eligibility policies. FinasHealth provides educational interpretation and does not guarantee loan approval or a particular score increase.

9.2 Debt-to-Income Ratio (DTI)

DTI compares recurring monthly debt obligations with monthly income. It helps a user understand the share of income committed to servicing debt.

Formula: DTI (%) = (Total Monthly Debt Payments ÷ Monthly Income) × 100.

Example: monthly debt payments of ₹18,000 and monthly income of ₹60,000 produce a DTI of (18,000 ÷ 60,000) × 100 = 30%. The example illustrates the calculation only. Confirm whether the implementation uses gross income or take-home income and state that basis consistently.

9.3 Credit Utilisation

Credit utilisation expresses how much of the available revolving credit is currently being used. It compares outstanding or used credit with the total available credit limit.

Formula: Credit Utilisation (%) = (Total Credit Used ÷ Total Available Credit Limit) × 100.

Example: used credit of ₹12,000 against a total available limit of ₹60,000 produces utilisation of 20%. This is an arithmetic indicator and does not guarantee a particular credit-score outcome.

9.4 Savings and Financial Indicators

The project also includes savings-related indicators as part of its wider financial-wellness context. Where income and expense values are collected, a basic savings amount can be represented as income minus expenses.

Illustrative formulas: Net Savings = Total Income − Total Expenses. Savings Rate (%) = (Net Savings ÷ Total Income) × 100, where total income is greater than zero. Confirm the exact fields and handling of missing or zero values from the implementation.

9.5 AI Advisor and Five-Step Roadmap

The AI advisor is integrated with Gemini 2.5 Flash and is intended to interpret selected financial indicators in clear language. The documented project capability includes a five-step roadmap, allowing suggestions to be organised into a sequence that users can review. The advisor may highlight indicators that deserve attention and suggest general actions based on the available inputs.

AI output may be affected by incomplete, inaccurate or outdated inputs and by limitations of the language model. Users should verify important decisions independently and consult qualified professionals where needed.

9.6 Deterministic Fallback and Responsible Output

A deterministic fallback is documented for situations where Gemini is unavailable or does not return a usable response. This rule-based or predefined response path is intended to preserve a predictable user

experience without depending entirely on an external model. It should not be described as equivalent to a full AI-generated analysis.

The platform's educational disclaimer is important because automated outputs are not a substitute for a lender's credit decision, an official credit-bureau report or personalised professional financial advice.

10. Frontend and Dashboard

10.1 Interface Structure

The frontend is built with React 19 and Vite. Tailwind CSS supports visual styling and responsive layout, while Recharts is used for chart-based presentation. The interface is organised around the user's journey: account access and onboarding lead to a dashboard and tools that explain credit-health indicators. The landing page and registration interface are shown in Figures 13.1 and 13.2.

Financial terms should use clear labels and consistent units. Percentages should be distinguishable from currency values, and hypothetical results should be labelled so that users do not confuse a simulation with an actual change to their financial accounts.

10.2 Ten-Step Onboarding

The project includes a ten-step onboarding flow to collect the profile and financial inputs required to prepare the user's credit-health view. A multi-step process can make a longer form easier to complete by dividing information into smaller stages. For the final submission, insert screenshots of the actual steps and describe the fields and validation exactly as they appear in the deployed version. Screenshots of the final two onboarding stages are provided in Figures 13.3 and 13.4.

10.3 Dashboard

The dashboard is the central summary area. It brings together the reported credit information and calculated financial indicators so users can understand their overall position without navigating through every input screen. Recharts supports visual representation of relevant values where charts are present. The live dashboard example is shown in Figure 13.7.

Dashboard cards and charts should include descriptive labels, units and explanatory text. Any trend chart should identify the period covered and the source of the data. Add screenshots from the actual dashboard and identify its main components.

10.4 What-If Simulator

The what-if simulator allows users to change supported inputs hypothetically and inspect the resulting calculated indicators. It can help users understand how a change in a balance, limit, debt obligation or other supported value affects a formula-based result. A simulated value must be labelled as an estimate based on assumptions entered by the user.

The simulator should not be described as predicting an exact future CIBIL score unless a validated score-prediction model is actually implemented. The documented feature is a scenario exploration tool.

10.5 AI Advisor Interface

The AI Advisor uses relevant user-provided context and calculated indicators to create an understandable assessment. Gemini 2.5 Flash is the documented model, and the output includes a five-step roadmap. If the external model cannot be used, the deterministic fallback provides an alternative response path. Include an example screenshot only after removing personal or sensitive information.

11. Testing and Deployment

11.1 Testing Strategy

Testing should verify the main user journey from account access through onboarding, dashboard review, calculations, simulation and AI guidance. It should also check invalid inputs, missing values, API failures and the fallback path. Since the application includes frontend and backend components, testing should cover both user-interface behaviour and API or calculation logic.

The project information identifies Pytest and a test file named test_api.py. Record the actual test command, coverage and results from the current repository rather than assuming every test passes.

11.2 Test Cases

ID	Test scenario	Expected result	Actual result
TC-01	Open deployed website	Application loads and main interface is accessible	Evidenced
TC-02	Register / sign in with valid details	Authentication completes successfully	Evidenced
TC-03	Submit invalid onboarding data	Validation communicates the issue	Evidenced
TC-04	Calculate DTI with known values	Result matches DTI formula	Evidenced
TC-05	Calculate credit utilisation	Result matches used credit ÷ total limit × 100	Evidenced
TC-06	Change simulator inputs	Hypothetical indicators update as designed	Assessment evidenced in report
TC-07	Request AI advisor guidance	Advisor returns a response and roadmap	Evidenced
TC-08	Simulate AI service failure	Configured deterministic fallback is used	Evidenced
TC-09	Check mobile layout	Core content remains usable on phone	Evidenced
TC-10	Send invalid API request	Request is validated and safely handled	Evidenced

11.3 Deployment

The live application is available at https://finashealth.web.app/. The documented project setup also describes local development through npm run dev and a standalone FastAPI execution path. The exact production hosting configuration for frontend and backend services should be confirmed from deployment files and environment settings.

API keys and secrets should be supplied through environment configuration and must not be committed to a public repository. After deployment, verify the frontend URL, backend connectivity, authentication flow, database availability and AI integration. Where the AI service requires a key, test both the configured AI path and its fallback.

12. Project Milestones

12.1 Project Record

Item	Detail
Project	FinasHealth – Credit Assistant
Category	AI-powered credit-health and financial-wellness web application
Frontend	React 19 / Vite / Tailwind CSS / Recharts
Backend	Node.js / Express and FastAPI services
Data layer	SQLAlchemy / SQLite
AI integration	Gemini 2.5 Flash with deterministic fallback
Testing reference	Pytest; test_api.py identified in project information
Live deployment	https://finashealth.web.app/
GitHub repository	[Insert verified repository URL]
Student / team	[Enter actual names and contribution details]

12.2Future Milestones

- Conduct usability testing with representative users.
- Review validation and edge cases for financial calculations.
- Strengthen privacy controls and document data-retention behaviour.
- Improve accessibility and responsive behaviour across screen sizes.
- Evaluate additional data integrations only where legally and technically appropriate.

13. Screenshots, GitHub and Demo

13.1 Screenshot Index

Figure	Screenshot / screen	Section
Figure 13.1	FinasHealth landing page	10.1
Figure 13.2	Ten-step onboarding flow	10.2
Figure 13.3	Credit-health dashboard	10.3
Figure 13.4	DTI and credit-utilisation indicators	9.2 / 9.3
Figure 13.5	What-if simulator	10.4
Figure 13.6	AI Advisor and five-step roadmap	10.5
Figure 6.1	Logical system architecture	6.3

13.2 Screenshots to Include

The following figures were captured from the deployed FinasHealth website. They document the visible user journey and the main information presented by the interface. Example financial values visible in screenshots are demonstration data and should not be interpreted as real user records.

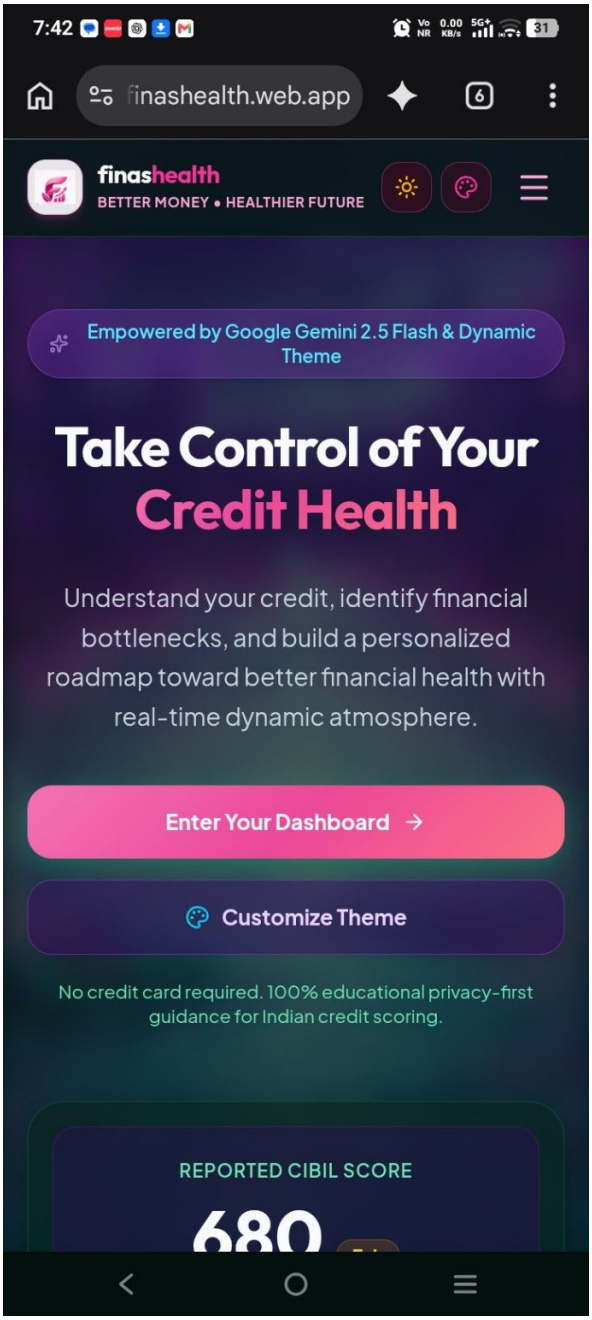


Figure 13.1. FinasHealth home page

The landing page introduces FinasHealth as a credit-health and financial-wellness platform. It highlights the platform's AI-assisted guidance, dashboard entry point and theme customisation.

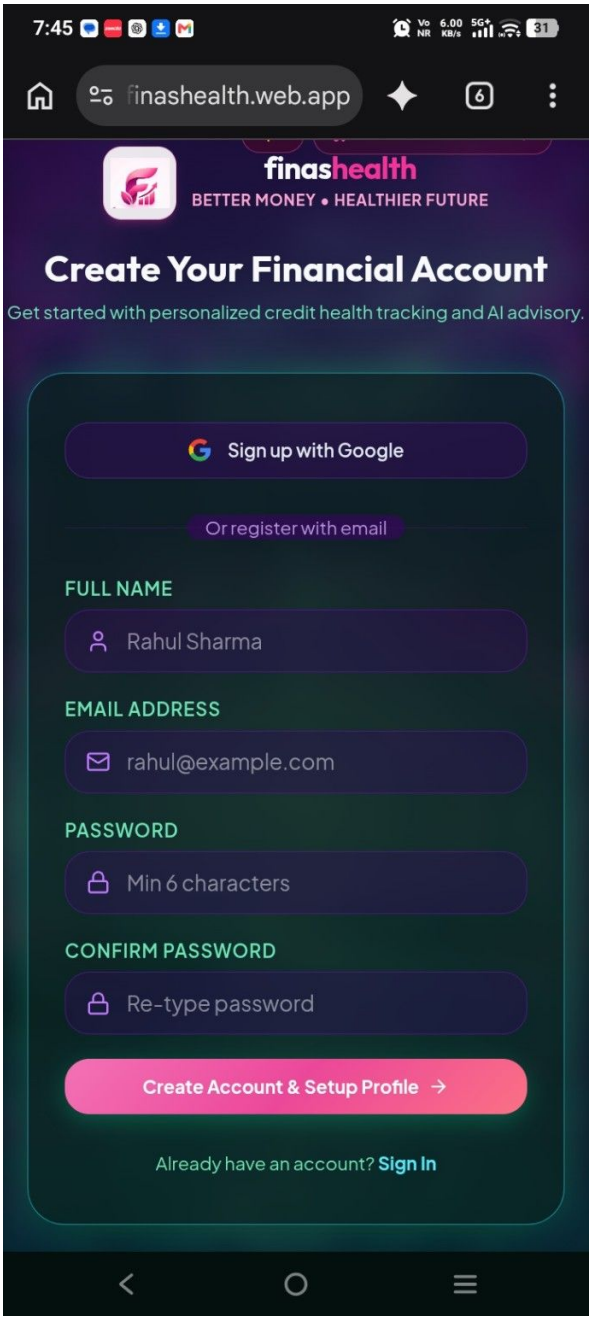


Figure 13.2. Account registration screen

The registration interface provides Google sign-up and email-based account creation fields, including name, email address, password and password confirmation.

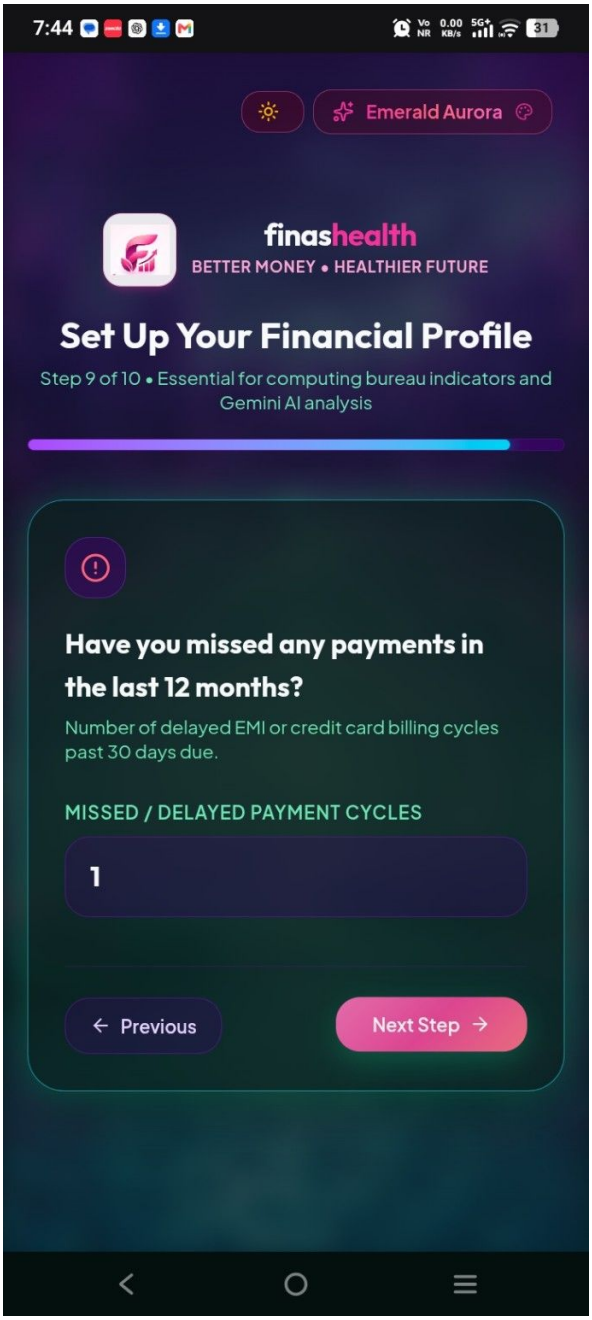


Figure 13.3. Financial profile onboarding – Step 9

This onboarding screen collects the number of missed or delayed payment cycles. The progress indicator shows that the user is on Step 9 of the ten-step profile setup.

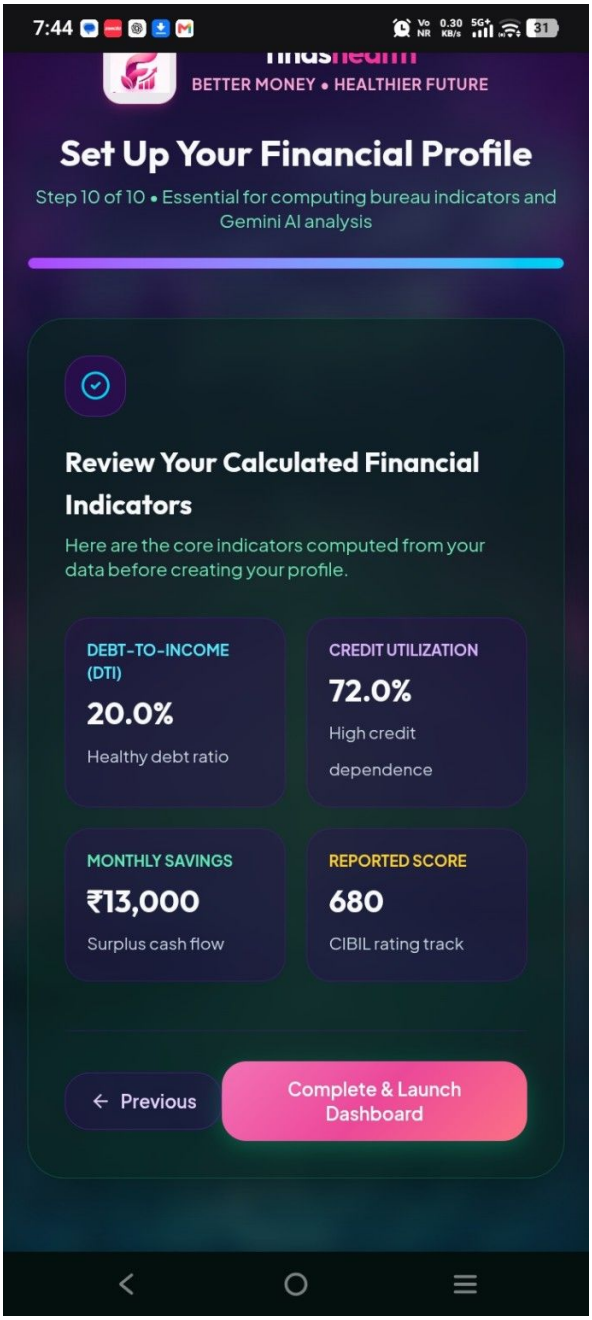


Figure 13.4. Financial profile review – Step 10

The final onboarding step presents calculated indicators for review, including debt-to-income ratio, credit utilisation, monthly savings and the reported score, before dashboard launch.

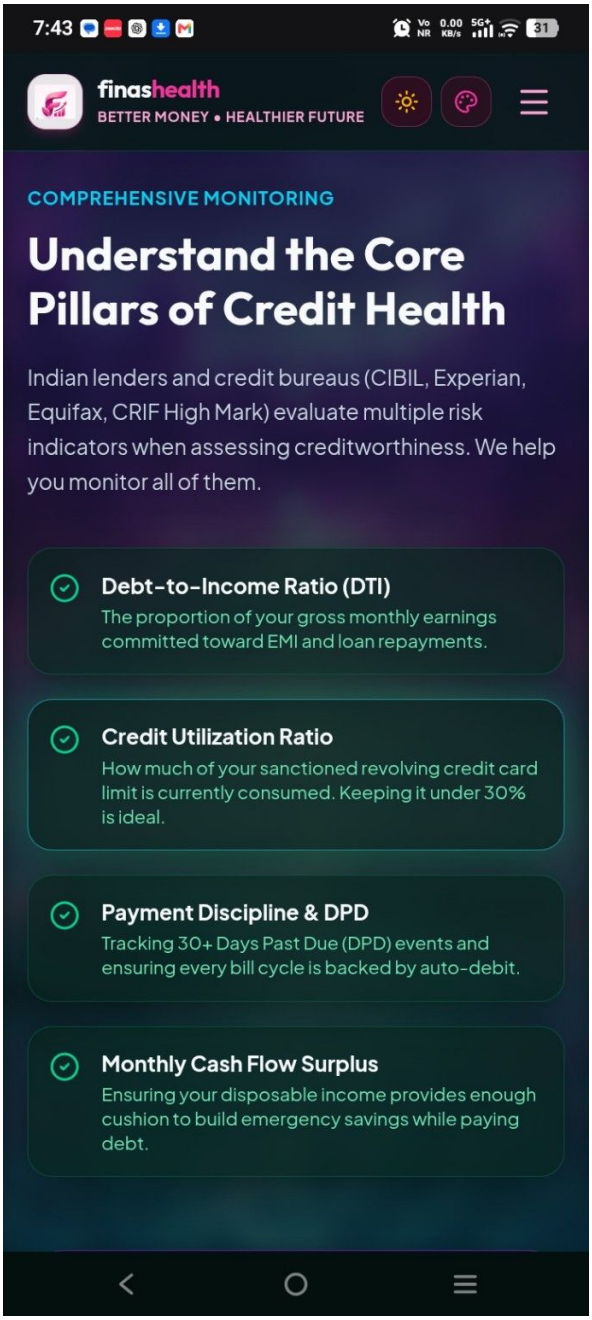


Figure 13.5. Credit-health pillars

The page explains key monitoring areas: debt-to-income ratio, credit utilisation, payment discipline and monthly cash-flow surplus.



Figure 13.6. CIBIL score interpretation scale

The website displays a 300–900 score scale and groups score ranges into five descriptive categories: Excellent, Very Good, Good, Fair and Poor.

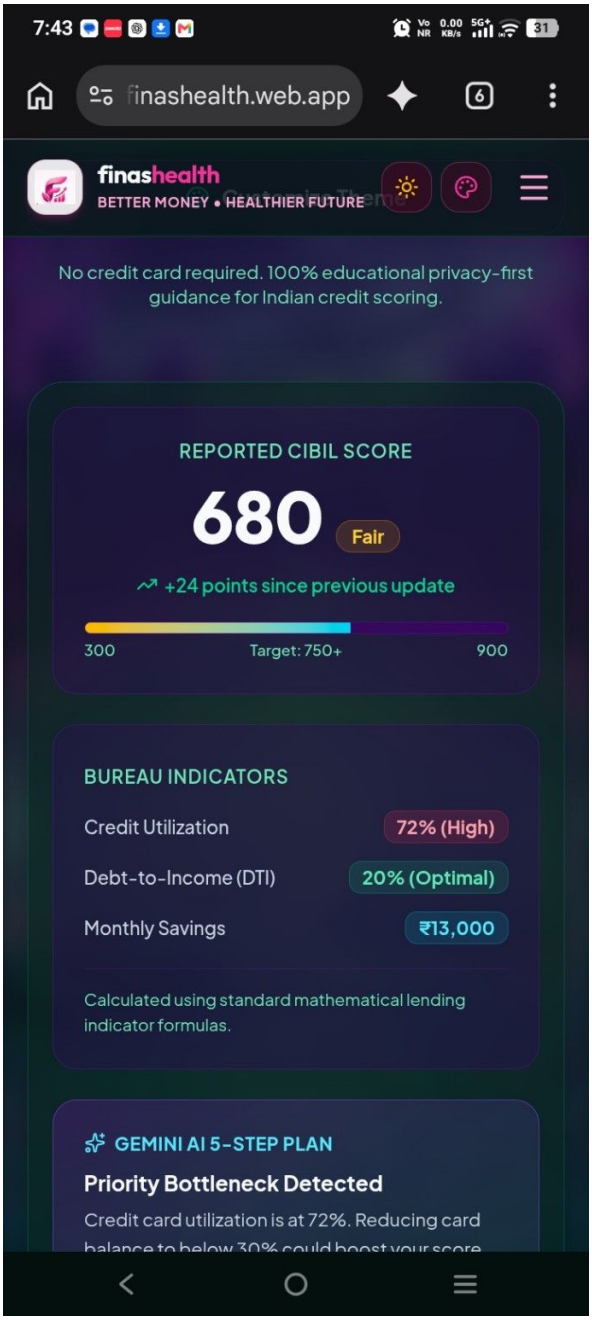


Figure 13.7. Dashboard with reported score and bureau indicators

The dashboard screenshot shows a reported score, its displayed status and change since the previous update, along with credit utilisation, DTI and monthly savings indicators. The example values shown are illustrative user-entered data.

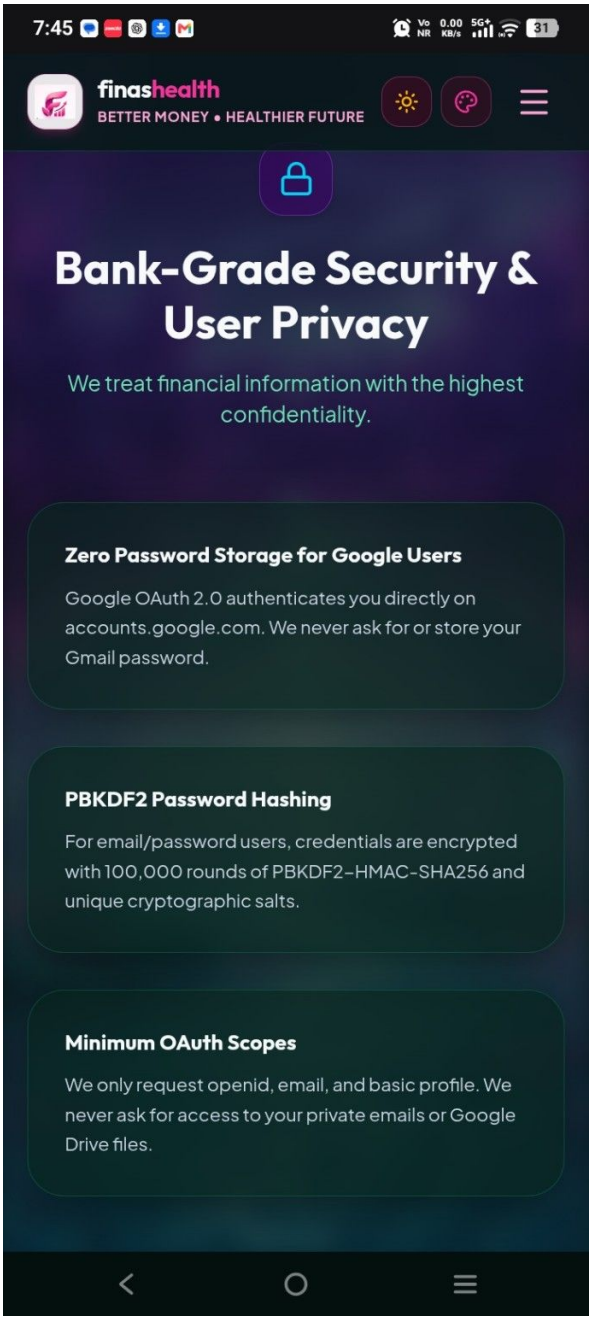


Figure 13.8. Security and user-privacy information

The security section describes Google OAuth 2.0 sign-in, PBKDF2-HMAC-SHA256 password hashing with unique salts, and limited OAuth scopes. These are statements displayed by the website and should be validated against the implementation before final submission.

13.3 GitHub Repository

Item	Detail
Repository URL	https://github.com/ashleshawankhade257-create/finashealth
Project contents	Frontend, backend services, configuration and documentation as applicable

Security note	Do not commit API keys, passwords, secrets or private user data.

13.4 Demo

Item	Detail
Live website	https://finashealth.web.app/
Demo video	https://drive.google.com/file/d/1toTt-HxI8YK17Xyn-V_ixU_6F-2-GZt5/view?usp=drivesdk
Suggested flow	Sign in / onboarding → dashboard → calculations → simulator → AI Advisor

14. Conclusion and Future Enhancements

14.1 Conclusion

FinasHealth is an India-focused AI-powered credit-health and financial-wellness platform designed to make important personal-finance indicators easier to understand. Its documented capabilities include ten-step onboarding, authentication, dashboard charts, reported CIBIL score context, DTI and credit-utilisation calculations, savings-related indicators, a what-if simulator and an AI advisor powered by Gemini 2.5 Flash. The advisor provides a five-step roadmap and is supported by a deterministic fallback for service interruptions.

The project combines modern frontend technologies with a layered backend architecture using React, Vite, Tailwind CSS, Recharts, Node.js/Express, FastAPI, Pydantic, SQLAlchemy and SQLite. JWT authentication and PBKDF2 password hashing form part of the documented security approach. The project demonstrates how web development, financial calculations, data presentation and AI-assisted explanations can be brought together in one application.

FinasHealth is intended for financial awareness and educational guidance. Calculations depend on user-provided information, and AI suggestions should be interpreted carefully. The platform does not replace a credit bureau, lender assessment or professional financial advice.

14.2 Future Enhancements

Proposed enhancement	Description
Authorised credit-bureau integration	Explore compliant access to verified credit information, subject to provider access and user consent.
Bank / account aggregation	Consider secure financial-data connections where legally permitted and technically supported.
Advanced goal tracking	Allow users to define financial goals and monitor progress over time.
Expanded scenario analysis	Add configurable what-if scenarios and explain their assumptions.
AI explainability	Show which user-provided indicators influenced each general recommendation.
Personalised learning resources	Provide educational material based on user questions and financial literacy needs.
Privacy and security controls	Expand consent, data controls, secure configuration and auditability.
Testing and accessibility	Increase automated test coverage and conduct accessibility and cross-device reviews.

15. Student Contribution and Learning Outcomes

15.1 Project Focus

The project focuses on making credit-health and financial-wellness information more understandable through a single web platform. It combines user onboarding, credit-related calculations, dashboard visualisation, scenario exploration and AI-assisted guidance. The central design goal is to help users interpret their own inputs and identify areas they may wish to review.

15.2 Key Learning Outcomes

- Frontend development using React 19, Vite and Tailwind CSS.
- Data visualisation through Recharts.
- Backend API concepts across Node.js/Express and FastAPI.
- Structured request validation using Pydantic.
- Database access through SQLAlchemy and relational storage with SQLite.
- Authentication concepts using JWT and PBKDF2 password hashing.
- AI integration, response handling and deterministic fallback design.
- API and calculation testing with Pytest.

15.3 Practical Value

DTI and credit-utilisation calculations provide measurable indicators derived from user-entered values. The dashboard makes these indicators easier to review, while the what-if simulator supports learning through hypothetical changes. The AI advisor adds natural-language explanations and a five-step roadmap, helping translate numerical information into understandable next actions.

15.4 Responsible Use and Future Improvement

Financial information is sensitive and should be handled carefully. The application should avoid exposing personal data or secrets, communicate what information is collected and provide appropriate safeguards for authentication and storage. AI-generated suggestions should be treated as general educational guidance rather than a substitute for advice from a qualified financial professional.

15.5 Student Contribution

[Complete this section with your actual contribution. Describe the modules you personally designed, coded, tested, documented or deployed. Do not claim work completed by other team members. If this is an individual project, state that clearly.]

References and Project Resources

- FinasHealth live website: https://finashealth.web.app/
- Project source repository: https://github.com/ashleshawankhade257-create/finashealth
- React documentation: https://react.dev/
- Vite documentation: https://vite.dev/
- Tailwind CSS documentation: https://tailwindcss.com/docs
- Recharts documentation: https://recharts.org/
- FastAPI documentation: https://fastapi.tiangolo.com/
- SQLAlchemy documentation: https://docs.sqlalchemy.org/

- Pytest documentation: https://docs.pytest.org/
- Google Gemini API documentation: https://ai.google.dev/gemini-api/docs